Supplementary Materials

Bridging Vision and Language in Construction: A Review of Automated Visual-to-Text Generation

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Table S1. Construction problems addressed by visual-to-text generation

Problem	Problem-Specific Task	Studies
Safety management	Hazard Identification and Risk Assessment	Wang et al. (2022), Zhai et al. (2023), Hsiao et al. (2024), Kim et al. (2024), Wang et al. (2024b), Zhu et al. (2024), Chen & Yin (2025), Guo et al. (2025), Guo et al. (2026), Sivanraj et al. (2026)
	Detection of Unsafe Behaviors and Non-Compliant Actions	Ding et al. (2022), Gil & Lee (2024), Abidi et al. (2025), Chan et al. (2025), Kim et al. (2025), Hussain et al. (2026), Wang et al. (2026), Xu et al. (2026)
	Ergonomic Risk Assessment	Fan et al. (2024), Yong et al. (2024a), Yong et al. (2024b), Yong et al. (2024c)
	Scene-Level Safety Understanding	Chen et al. (2023a), Zhong et al. (2023)
	Safety-Inspection Reporting	Tsai et al. (2023), Chen et al. (2024), Tsai et al. (2025)
Progress monitoring	Resource & Location-Based Progress Tracking	Bang & Kim (2020)
	work activity log	Xiao et al. (2024)
Scene documentation	Construction Activity Documentation	Liu et al. (2020), Ding & Luo (2021), Xiao et al. (2022), Song et al. (2024), Deng et al. (2025) ; Jung et al. (2024),
	Infrastructure Inspection Documentation	Pu et al. (2024a), Pu et al. (2024b), Chang et al. (2024), Bazrafshan et al. (2025)
	Scene-level environment understanding	Chen et al. (2023b), Wang et al. (2024a), Sree et al. (2025)
Damage & condition assessment	Disaster damage assessment	Aydin et al. (2026)
	Structural distress / defect inspection	Dinh et al. Ahn (2024); Watanabe et al. (2024)

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Table S2. Visual data acquisition platforms used in the reviewed studies

Note: If a study incorporates more than one acquisition platform, it is assigned to all relevant categories.

Main category	Acquisition Platform	Studies
Aerial-level	UAV / drone	Bang & Kim (2020); Xiao et al. (2022); Abidi et al. (2025); Aydin et al. (2026); Xu et al. (2026).
	Satellite imagery	Chen et al. (2023b); Wang et al. (2024a); Sree et al. (2025).
Ground-level	Fixed cameras (CCTV / surveillance)	Xiao et al. (2022); Zhong et al. (2023); Chen et al. (2024); Hsiao et al. (2024); Xiao et al. (2024); Chan et al. (2025); Chen & Yin (2025); Sivanraj et al. (2026).
	Manual photos (inspector-taken / mobile or camera)	Liu et al. (2020); Ding & Luo (2021); Wang et al. (2022); Chen et al. (2023a); Tsai et al. (2023); Zhai et al. (2023); Chang et al. (2024); Dinh et al. Ahn (2024); Gil & Lee (2024); Jung et al. (2024); Kim et al. (2024); Pu et al. (2024b); Song et al. (2024); Wang et al. (2024b); Watanabe et al. (2024); Zhu et al. (2024); Bazrafshan et al. (2025); Chen & Yin (2025); Deng et al. (2025); Guo et al. (2025); Kim et al. (2025); Tsai et al. (2025); Guo et al. (2026); Hussain et al. (2026).
	Wearables (AR helmets/body cams)	Chen et al. (2024)
	Robotic platforms (UGV / quadruped robots)	Pu et al. (2024a); Abidi et al. (2025)
	Web-based (internet-crawled / YouTube / istock)	Ding & Luo (2021); Ding et al. (2022); Xiao et al. (2022); Zhong et al. (2023); Zhai et al. (2023); Fan et al. (2024); Yong et al. (2024a); Yong et al. (2024b); Yong et al. (2024c); Abidi et al. (2025); Bazrafshan et al. (2025); Chan et al. (2025); Chen & Yin (2025); Deng et al. (2025); Guo et al. (2025); Hussain et al. (2026); Wang et al. (2026); Xu et al. (2026).
	Synthetic images	Xu et al. (2026)

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Table S3. Annotation classes across datasets

Note: Reported only if specified by the source.

Dataset	Classes
MOCS (An et al., 2021)	13 construction object classes Worker; Tower crane; Hanging hook; Vehicle crane; Roller; Bulldozer; Excavator; Truck; Loader; Pump truck; Concrete mixer truck; Pile driver; Other vehicle
ACID (Xiao & Kang, 2021)	10 construction machine classes Excavator; Compactor; Bulldozer; Grader; Dump truck; Concrete mixer truck; Wheel loader; Backhoe loader; Tower crane; Mobile crane
SODA (Duan et al., 2022)	16 construction operation scene categories Tower crane lifting; Truck crane lifting; Crawler crane lifting; Gantry crane lifting; Hoisting steel components; Hoisting prefabricated components; Hoisting building materials; Equipment installation lifting; Structural assembly lifting; Material transportation lifting; Multi-worker cooperative lifting; Ground guiding operations; Suspended load positioning; Load binding operations; Site lifting preparation; General lifting workflow scenes
Construction Scene Graph Dataset (Zhang et al., 2022)	9 construction object categories: Worker; Hardhat; Brick; Welder; Facial mask; Glove; Scaffold; Vest; Excavator 7 interaction types: Drive; Under; Lay; Stand on; Operate; Wear; Next to
CPPED (Chen et al., 2025)	13 PPE-related object classes: person; head; safety vest; safety glasses; mask; gloves; no-gloves; shoes; no-shoes; safety helmet (white, blue, red, yellow)
Harvard Dataverse Hardhat Dataset (Xie, 2019)	Hardhat presence/absence No additional information is specified.
Ladder dataset (Anjum et al., 2022)	5 ladder safety classes: Ladder with outriggers; Ladder without outriggers; Worker with helmet; Worker without helmet; Worker with safety belt
Mobile scaffold dataset (Khan et al., 2021)	3 construction safety classes: Person; Mobile scaffold with outriggers (safescaffold); Mobile scaffold without outriggers (scaffold)
iSafe Welding System dataset (Zaidi et al., 2023)	5 welding safety classes: Welding equipment; Worker with PPE; Welding machine; Fire extinguisher; Flammable material
FloodNet (Rahnemoonfar et al., 2021)	9 semantic classes: building-flooded; building-non-flooded; road-flooded; road-non-flooded; water; tree; vehicle; pool; grass
LADI-v2 (Scheele et al., 2024)	12 damage and infrastructure classes: Bridges_any; Buildings_any; Buildings_affected_or_greater; Buildings_minor_or_greater; Debris_any; Flooding_any; Flooding_structures; Roads_any; Roads_damage; Trees_any; Trees_damage; Water_any

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Table S4. Model architecture of reviewed studies

Model Architecture	Core Backbone	Studies
Classical Encoder-Decoder	CNN + LSTM	Bang & Kim (2020); Liu et al. (2020); Wang et al. (2022); Chen et al. (2023b); Zhai et al. (2023); Zhong et al. (2023); Dinh et al. Ahn (2024); Zhu et al. (2024); Wang et al. (2024b); Sree et al. (2025); Deng et al. (2025); Wang et al. (2026)
	CNN + Transformer	Xiao et al. (2022); Chen et al. (2023a)
	CNN + GRU	Hsiao et al. (2024)
	Swin Transformer + Transformer	Song et al. (2024)
Hybrid Models	CLIP + GPT-2	Tsai et al. (2023); Gil & Lee (2024); Tsai et al. (2025); Guo et al. (2026)
	CLIP + LLaMA	Wang et al. (2024a)
	CNN + GPT-3.5	Xiao et al. (2024)
	CNN + GPT-4	Hussain et al. (2026)
Vision-Language Pretrained Models	UNIMO	Ding & Luo (2021)
	ViLT	Ding et al. (2022)
	OTTER	Watanabe et al. (2024)
	InstructBLIP	Yong et al. (2024c)
	mPLUG	Jung et al. (2024)
	BLIP + BART	Yong et al. (2024b)
	BLIP	Chen et al. (2024)
	BLIP-2	Abidi et al. (2025); Aydin et al. (2026)
	Florence-2	Sivanraj et al. (2026)
Large Multi-Modal Models	MiniGPT-4	Pu et al. (2024a); Chang et al. (2024); Pu et al. (2024b); Yong et al. (2024a)
	LLaVA	Kim et al. (2024)
	GPT-4V	Fan et al. (2024); Kim et al. (2025); Bazrafshan et al. (2025)
	CogVLM-17B	Chan et al. (2025)
	InternVL-2 Pro	Guo et al. (2025)
	Qwen2-VL-7B	Chen & Yin (2025)
	Qwen2.5-VL-7B	Xu et al. (2026)

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Table S5. Evaluation metrics for construction visual-to-text generation

Metric	**Equation**
BLEU	$\mathrm{BLEU} = \exp\left(\sum_{n=1}^{N} w_n \log p_n\right).min\left(1, e^{1-r/c}\right)$
ROUGE	$\text{ROUGE-L} = \frac{(1+\beta^2)R_{LCS} \cdot P_{LCS}}{R_{LCS} + \beta^2 \cdot P_{LCS}}$ $\text{ROUGE-N} = \frac{(1+\beta^2)R_N \cdot P_N}{R_N + \beta^2 \cdot P_N}$
METEOR	$\mathrm{METEOR} = (1 - \mathrm{Pen}) \cdot F_{mean}$ $F_{mean} = \frac{P.R}{\alpha.P + (1-\alpha).R}$
CIDEr	$\mathrm{CIDEr}(C_i, S_i) = \sum_{n=1}^{N} \omega_n \, \mathrm{CIDEr}_n(C_i, S_i)$ $\mathrm{CIDEr}_n(C_i, S_i) = \frac{1}{m}\sum_j cos(C_i, s_{ij})$
BERSTScore	$F_{BERT} = 2 \cdot \frac{P_{BERT} \cdot R_{BERT}}{P_{BERT} + R_{BERT}}$ $R_{BERT} = \frac{1}{\lvert S\rvert}\sum_{j\in S} \max_{i\in C} \cos\,(S_j, C_i)$ $P_{BERT} = \frac{1}{\lvert C\rvert}\sum_{i\in C} \max_{j\in S} \cos\,(S_i, C_j)$
WMD	$\mathrm{WMD}(S, C) = \min_{T\geq 0} \sum_{i,j=1} T_{ij} \,.\, \Delta(.,.)$ $\sum_j T_{ij} = d_i, \sum_i T_{ij} = d'_j$
Cosine sim	$\cos(C, S) = \frac{C \cdot S}{\lVert C\rVert \lVert S\rVert}$
SPICE	$\mathrm{SPICE} = \frac{2 \cdot P(C,S) \cdot R(C,S)}{P(C,S) + R(C,S)}$
Keyword recall	$\text{Keyword Recall(C,S)} = \frac{\lvert K(S) \cap K(C)\rvert}{\lvert K(C)\rvert}$
CLIPScore	$\mathrm{CLIPScore} = W.\max(\cos(C, V), 0)$
Recall@K	$\text{Recall@K} = \frac{\left\lvert\{q \in Q: S_q \in top - K(q)\}\right\rvert}{\lvert Q\rvert}$
IoU	$\mathrm{IoU} = \frac{TP}{\mathrm{TP+FN+FP}}$
AP	$\mathrm{AP} = \sum_k P(k).\Delta R(k)$
mAP	$\mathrm{mAP} = \frac{1}{\mathcal{C}}\sum_{\varsigma\in\mathcal{C}} \mathrm{AP}_{\varsigma}$
Accuracy	$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$
Precision	$\mathrm{Precision} = \frac{TP}{TP + FP}$
Recall	$\mathrm{Recall} = \frac{TP}{TP + FN}$
F1-score	$\mathrm{F1} = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$
F2-score	$\mathrm{F2} = \frac{(1 + 2^2) \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{2^2 \cdot \mathrm{Precision} + \mathrm{Recall}}$

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Table S6. Notation of evaluation metrics

Variable	Definition
AP_{ς}	Average Precision for category ς
$CIDEr_n(c_i, S_i)$	Average cosine similarity between candidate and reference
f_i	Normalized frequency of word *i* in candidates *C*
f'_j	Normalized frequency of word *j* in references *S*
P_{LCS}	Precision based on the longest common subsequence
p_n	Modified n-gram precision for order *n*
P_N	*n-gram* recall - fraction of reference *n-grams* matched in candidate *C*
R_{LCS}	Recall based on the longest common subsequence
R_N	*n-gram* precision – fraction of candidate *n-grams* matched in reference *S*
S_q	Correct candidate for query *q*
T_{ij}	Transport flow from word *i* in candidate *C* to word *j* in reference *S*
w_n	Weight for *n-grams*
ω_n	Uniform weight per *n-gram* order
$\Delta(.,.)$	Euclidean distance
c	Candidate length
C	Set of candidate token embeddings
C, S	Candidate and reference document
$cos(.,.)$	Cosine similarity between two token embeddings
F_{mean}	Harmonic mean
FN	False negative
FP	False positive
K	Number of top retrieved results considered
K(C)	Set of domain-specific keywords found in candidate *C*
K(S)	Set of all relevant domain-specific keywords in reference *S*
N	*n-gram* order
n	*n-gram* order
P	Precision
P(C, S)	Precision of candidate *C* and reference *S*
$P(k)$	Precision at threshold *k*
Pen	Fragmentation penalty
q	A single query
Q	Set of all queries
r	Reference length
R	Recall
R(C, S)	Recall of candidate *C* and reference *S*
$R(k)$	Change in recall at threshold k
S	Set of reference token embeddings
TN	True negative
$top-K(q)$	Set of top *K* retrieved candidates for query *q*
TP	True positive
V	Clip the visual embedding of the input image
α	Weight factor in the harmonic mean
Β	Recall/precision balance weight
$\mathcal{C}$	Set of all object categories
ς	A single object category
ς	A single object category

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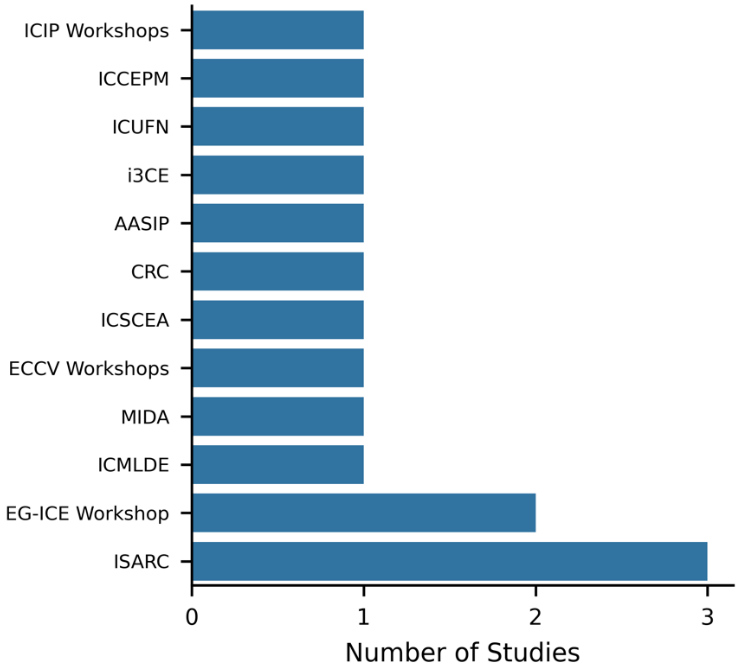

Fig. S1. Distribution of Identified Studies Across Conference Venues
Abbreviations:
ICIP = IEEE International Conference on Image Processing
ICCEPM = International Conference on Construction Engineering and Project Management
ICUFN = International Conference on Ubiquitous and Future Networks
i3CE = ASCE International Conference on Computing in Civil Engineering
AASIP = International Conference on Advanced Algorithms and Signal Image Processing
CRC = Construction Research Congress
ICSCEA = International Conference on Sustainable Civil Engineering and Architecture
ECCV = European Conference on Computer Vision
MIDA = International Conference on Machine Intelligence and Digital Applications
ICMLDE = International Conference on Machine Learning and Data Engineering
EG-ICE = European Group for Intelligent Computing in Engineering
ISARC = International Symposium on Automation and Robotics in Construction

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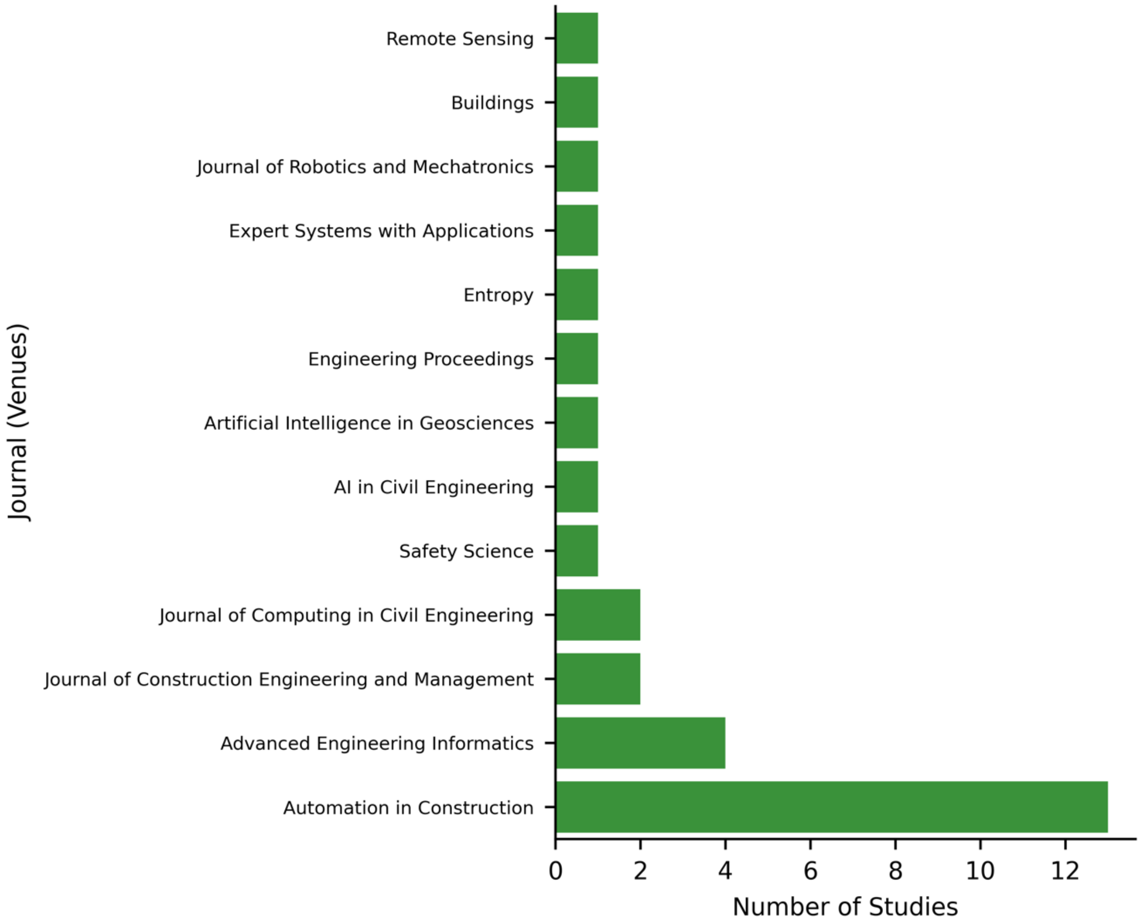

Fig. S2. Distribution of identified studies across journal venues
Publishers and their journals: Elsevier (Automation in Construction, Advanced Engineering Informatics, Expert Systems with Applications, Artificial Intelligence in Geosciences, Safety Science); ASCE (Journal of Construction Engineering and Management, Journal of Computing in Civil Engineering); MDPI (Buildings, Entropy, Engineering Proceedings, Remote Sensing); Springer (AI in Civil Engineering); Fuji Technology Press (Journal of Robotics and Mechatronics).